

Progress Report 1 – Initial Stage

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1. Student Details

Name	Md Faiz Alam
Course	BCA (Bachelor of Computer Applications)
Semester	5 th Semester
Admission No.	UGO24156735
Project Title	QuizMind AI – An Adaptive Quiz platform for Interview & Placement

2. Project Objectives

QuizMind AI is a web-based adaptive quiz platform designed for computer science and data science students preparing for technical interviews and campus placements. The system uses an intelligent quiz engine that adjusts question difficulty based on user performance, tracks weak areas at the subtopic level, and provides AI-powered explanations and personalized study recommendations — all within a single, freely accessible platform

3. Proposed Features

- User registration and login system with secure password hashing
- Topic and subtopic selection across 6 domains: Python, SQL, DSA, Machine Learning, Statistics, Data Analysis
- Adaptive quiz engine with 894+ real interview-style questions across 35 subtopics
- Countdown timer per question with automatic progression
- Instant AI-powered explanation shown after each wrong answer
- Performance dashboard showing accuracy, quizzes taken, and topics covered
- Weak area detection — identifies subtopics below 60% accuracy
- Quiz history and result saving with score, accuracy, and time taken
- Analytics page with bar chart visualization using Recharts
- Personalized AI study suggestions based on user's weak areas
- Cloud database using Supabase (PostgreSQL) accessible from anywhere
- Fully deployed web application — accessible from any device globally

4. Technology Stack

Category	Technology	Purpose
Frontend Language	JavaScripts (React.js + Vite)	User interface and quiz interaction

Backend Language	Python (Flask)	REST API and business logic
Database	Supabase (PostgreSQL)	Cloud database – questions, user, results
Authentication	Custom (SHA-256 hashing)	Secure user login and registration
Analytics	Recharts	Performance visualization
Hosting – Frontend	Vercel	Free frontend deployment
Hosting – Backend	Render.com	Free backend deployment
Version Control	Git + GitHub	Source code management

5. System Design

The system follows a client-server architecture with three layers:

- Frontend Module (React.js): Five main pages — Login/Register, Dashboard, Quiz Screen, Results, and Analytics. The user selects a topic and subtopic, attempts the quiz with a live timer, and sees results with AI suggestions.
- Backend Module (Flask API): RESTful API with three route groups — /api/auth (register, login), /api/quiz (fetch questions by subtopic), and /api/results (save results, fetch analytics, detect weak areas).
- Database Module (Supabase PostgreSQL): Six tables — topics, subtopics, questions, users, results, and topic_scores. Questions are pre-seeded with 894 real interview-style MCQs across 35 subtopics.
- Data Flow: User logs in -> selects subtopic -> frontend fetches questions from Flask API -> user attempts quiz -> results saved to Supabase -> analytics dashboard fetches performance data and displays charts.

6. Work Done So Far dgg

- Project idea finalized and problem statement defined
- System architecture and technology stack decided
- GitHub repository created and initial project structure set up
- React.js frontend project initialized using vite
- Flask backend created with basic folder structure and route files
- Cloud database configured on Supabase with all required tables created
- User login and registration system implemented
- Basic quiz screen UI started